Novelty-Driven Mean Reversion

A Portfolio Sort Study of Earnings Conference Call Text Analytics

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1. Introduction

Earnings conference calls are among the most information-rich corporate disclosure events. Beyond the headline numbers, management's prepared remarks and responses to analyst questions contain qualitative signals about strategy, risk, and forward guidance that are difficult to extract from structured data alone. A growing literature in textual analysis of financial disclosures has demonstrated that linguistic features such as tone, sentiment, complexity, and similarity, carry incremental information content beyond what is captured by quantitative earnings surprises.

This study introduces a novel measure of *semantic novelty* wihch is the degree a firm's earnings call transcript is linguistically distant from its own historical baseline, and tests whether this measure predicts abnormal returns and subsequent mean reversion. The central intuition is straightforward: when a firm discusses topics it has never discussed before (an AI partnership by a legacy manufacturer, a new regulatory risk, a pivot in business model), markets must process genuinely new information with limited precedent. Behavioral finance predicts that investors will overreact to salient, novel stimuli, and that the initial mispricing will partially correct as sophisticated traders arbitrage the excess reaction.

Our contribution is threefold. First, we develop a multi-layered NLP pipeline that goes beyond simple sentiment or bag-of-words approaches by computing turn-level semantic novelty using transformer encoders, filtering for financial salience, penalizing systemic macro topics, and weighting by section type and analyst attention. We use `nomic-embed-text-v1.5` as our primary semantic encoder that is a purpose-built embedding model trained for retrieval and semantic similarity tasks and a locally-hosted LLM via Ollama as a downstream intent filter to verify whether flagged novelty represents a genuine business catalyst rather than noise or disclosure boilerplate. Second, we address the problem of *pre-priced novelty* which is the possibility that information flagged as novel in the transcript was already disclosed via press releases or news between earnings calls through a pre-event drift control and a topic depth filter. Third, we benchmark our portfolio returns directly against the methodology of Cohen, Malloy & Nguyen (2020), who construct a long-short portfolio on 10-K language changes and report up to 188 bps/month in alpha; our design extends their framework to earnings call transcripts, higher-frequency semantic distance measurement, and a cross-sectional macro filter absent from their approach.

The remainder of this memo proceeds as follows. Section 2 states the research question and hypotheses. Sections 3 and 4 describe the data and governance procedures. Section 5 presents exploratory data analysis. Section 6 details the portfolio construction methodology. Section 7 specifies the regression models.

2. Research Question & Hypotheses

Research Question

Do firms whose earnings call transcripts contain semantically novel and financially salient content relative to their own historical linguistic baseline, generating predictable cross-sectional differences in risk-adjusted returns, and does a long-short portfolio that exploits this signal earn significant alpha after controlling for standard risk factors and earnings surprises?

Hypotheses

H1 (Overreaction). A long-short portfolio that buys firms in the bottom quintile of adjusted novelty scores (Q1, familiar language) and shorts firms in the top quintile (Q5, novel language) earns positive risk-adjusted alpha, consistent with the market overpricing novel disclosures relative to familiar ones. We benchmark this alpha against Cohen, Malloy & Nguyen (2020), who report up to 188 bps/month from a long-short strategy on 10-K language changes.

H2 (Mean Reversion). High-novelty firms (Q5) exhibit larger initial abnormal returns around the earnings call date, followed by partial reversal in the post-event window [+2, +5], consistent with overreaction to semantically unfamiliar

content and subsequent correction by sophisticated investors.

H3 (Interaction with Sentiment). The overreaction effect is amplified when novel content carries positive FinBERT-scored tone, such as new projects, partnerships, and product launches framed optimistically, as retail and algorithmic traders disproportionately chase positive novelty. The long-short alpha is expected to be largest in the double-sorted portfolio that isolates high-novelty, high-positive-tone calls.

Mechanism

Markets are efficient at pricing well-understood, recurring information but overreact to genuinely novel disclosures. When a firm discusses something semantically distant from its historical baseline (for example, a legacy manufacturer announcing an AI partnership or a retailer pivoting to subscription revenue), sentiment-driven traders and algo models lack a reference frame and amplify the initial price move. Sophisticated investors subsequently correct the mispricing, generating the mean reversion that the long-short portfolio captures. The salience filter ensures we isolate financially material novelty (guidance changes, new products, capital allocation shifts) from noise (boilerplate, tangential remarks, regulatory disclosures). The analyst attention measure provides a revealed-preference signal from sophisticated market participants about which novel topics are genuinely surprising versus already widely known.

3. Data Sources & Data Definitions

3.1 Financial Data

The dataset comprises 202,357 earnings conference call events spanning April 2005 through December 2025, covering 6,867 unique tickers. All events are classified as "Earnings Calls." Financial data includes:

- **Daily returns:** Pre-computed simple daily returns from `ret_t-15` through `ret_t+15`, providing a 31-trading-day window centered on each event date. For the portfolio sort method, we will use a three month holding period from `t=0`.
- **Overnight return:** `close_to_open_return`, computed from `close_price_call_day` and `open_price_next_day`, serving as a proxy for the market's immediate earnings surprise reaction.
- **Event dates:** `mostimportantdateutc` identifies the earnings call date; `actual_call_date` is used for alignment.
- **Identifiers:** CRSP `permno` links each observation to the CRSP universe, enabling standard risk adjustment. Approximately 5,093 observations have missing permno values.

3.2 Text Data

Each observation includes the full earnings call transcript in the `full_transcript_text` column. Transcripts are the primary input for the NLP pipeline.

- **Preprocessing:** Operator instructions and safe-harbour boilerplate are removed via regular expressions. No stemming or lemmatization is applied, as transformer embeddings operate on raw text and benefit from intact morphological information.
- **Chunking:** Rather than splitting transcripts into fixed 200-word chunks, we embed at the speaking turn level, producing one embedding per component row. This is the natural unit of the data: each turn represents a

semantically coherent utterance from a single speaker, making artificial word-boundary splits unnecessary and counterproductive. Turn-level embeddings also preserve speaker identity directly, enabling section-level and role-level weighting (prepared vs. Q&A, management vs. analyst) without heuristic regex splitting. Turns exceeding the encoder's context window (~512 tokens for nomic-embed-text-v1.5) are split at sentence boundaries and their embeddings averaged, but this applies to fewer than 5% of turns in practice given typical speaking turn lengths. The 50-word overlap logic is dropped entirely as it was designed solely to mitigate boundary loss in the chunking approach.

- **Section splitting:** Prepared remarks and Q&A sections are identified directly from the transcriptcomponenttypename or equivalent section field in the component data, replacing the regex-based heuristic previously used on the raw transcript string.

3.3 Constructed Variables

Novelty Score (per turn). Semantic distance of each text turn from the firm's historical linguistic centroid:

$$\text{novelty} = 1 - \cos\big(\text{embed}(\text{turn}),\ \text{centroid}_{\text{hist}}\big)$$

where the historical centroid is the mean embedding of all turns from a rolling window of the firm's prior transcripts (strictly before the event date). The primary encoder is `nomic-embed-text-v1.5`, a 768-dimensional model trained on large-scale contrastive retrieval tasks and well-suited for semantic similarity comparisons.

Rolling Baseline Window. To prevent the historical centroid from being diluted by language from structurally different business periods, the baseline is computed using a rolling window rather than the full history. Two rolling specifications are tested: (a) all prior transcripts with exponential decay weighting (recent quarters weighted more heavily), and (b) a fixed trailing four-quarter window. Firms with fewer than two prior transcripts within the window are excluded.

Lexical Novelty Alternatives (TF-IDF and BM25). In addition to the semantic embedding approach, two lexical baselines are computed for direct comparison:

- *Firm-level rolling TF-IDF:* A TF-IDF vector is fitted on each firm's trailing transcripts. New-quarter turns are scored against this per-firm vocabulary. This is the closest analog to Cohen et al.'s cosine TF-IDF similarity measure and serves as the primary lexical baseline.
- *Trailing three-month universe TF-IDF:* A TF-IDF model fitted on all firms' transcripts from the prior three months, so novelty is measured relative to the current cross-sectional language environment rather than the firm's own history. This captures topics that are new to the broader market conversation, not just the individual firm.
- *BM25 lexical score:* A rolling per-firm-quarter BM25 score (k_1 = 1.5, b = 0.75) provides a third lexical benchmark with built-in document-length normalization and TF saturation.

These lexical alternatives are used in robustness tests: if the semantic embedding novelty score significantly outperforms the TF-IDF and BM25 versions in predicting returns, that is direct evidence that transformer-based semantic distance adds value beyond vocabulary change, which is the key methodological contribution over Cohen et al.

LLM Intent Filter (per high-novelty turn). High-novelty, high-salience turns (novelty ≥ 0.60, salience ≥ 0.30) are passed to a locally-hosted LLM via Ollama for binary intent classification. The prompt asks whether the turn describes a definitively new revenue-generating project, product, or corporate contract, returning YES or NO. This filter removes genuine linguistic novelty that is not a business catalyst: a new lawsuit, a CEO tangent, or a regulatory disclosure may

all register as semantically novel but do not represent the type of information expected to generate overreaction. Only turns passing the LLM filter contribute to the adjusted novelty score. The Ollama approach eliminates API cost and latency concerns and allows the filter model to be swapped without pipeline changes.

> Prompt: You are a financial analyst reviewing a sentence from an earnings call transcript. Sentence: "{sentence}" Does this sentence contain information that is likely to be financially material and meaningfully different from routine earnings call language? This includes, but is not limited to, new strategic initiatives, product or service changes, partnerships, contracts, market expansion, pricing decisions, cost actions, management changes, capital allocation shifts, or any forward-looking statement that represents a departure from prior guidance or company narrative. Exclude only clear boilerplate (safe-harbour disclaimers, operator instructions, generic pleasantries) and retrospective descriptions of already-reported results with no forward-looking content. Reply with ONLY one word: YES or NO.

Standardized Unexpected Earnings (SUE). Earnings surprise relative to analyst consensus:

$$\mathrm{SUE}_{i,t} = \frac{\mathrm{EPS}_{i,t}^{\mathrm{actual}} - \mathrm{EPS}_{i,t}^{\mathrm{consensus}}}{\sigma(\text{analyst forecasts})_{i,t}}$$

Sourced from I/B/E/S. SUE is a critical control in all regression specifications: a high-novelty call may simply be one where the firm dramatically beat earnings, producing novel language because the results themselves are novel. Without SUE, it is impossible to separate the language signal from the fundamental signal. The `close_to_open` return is an imperfect substitute because it conflates the market's reaction to both the numbers and the language simultaneously.

Salience Score (per turn). Financial materiality scored via a weighted 26-cluster dictionary (`salience_dictionary.py`). The score is the weighted sum of matched terms across clusters covering guidance, earnings, revenue, margins, and other financially material topics. See Appendix A for cluster details and sample terms.

Cross-Market Frequency Penalty. Topics appearing in a significant fraction of firms in the same quarter receive a discount reflecting their systemic rather than idiosyncratic nature. Cross-market frequency is computed using a rolling three-month trailing window of prior firm transcripts to avoid look-ahead bias: a topic's frequency is measured from what was already known at the time of the call, not from contemporaneous filings that may not yet have occurred. Rather than a linear penalty with a hard cliff, we apply a sigmoid decay that smoothly transitions from full weight at low cross-market frequency to near-zero weight at high frequency:

$$\mathrm{penalty}(f) = 1 - \frac{1}{1 + e^{-k(f - f_0)}}$$

where f is the fraction of firms in the trailing three-month window for which the topic cluster is the dominant novelty cluster, $f_0 = 0.40$ is the inflection point, and $k = 15$ controls the steepness of the transition. Under this specification, a topic appearing in 40% of prior-window firms retains a weight of approximately 0.50, a topic appearing in 60% retains approximately 0.07, and a topic appearing in fewer than 20% is penalised by less than 5%. This eliminates the discontinuity at the 40% threshold and reduces sensitivity to the precise threshold choice. The inflection point f_0 and steepness k are treated as hyperparameters and tested across a grid prior to finalising the specification.

Section Weight (per turn). Prepared remarks receive a weight of 1.0; Q&A management responses receive 1.5. The rationale is that Q&A responses are less scripted and elicited by analyst pressure, making novelty in this section a stronger signal of material information.

Analyst Attention Score (per topic cluster). Q&A turns are clustered by cosine similarity (threshold 0.75) into topic groups. Each cluster's attention score is the fraction of unique analysts who asked about that topic. Topics that multiple analysts probe receive higher weight; single-analyst tangents are downweighted.

Topic Depth (per topic per firm-event). The fraction of transcript turns devoted to each topic cluster, normalized by the median and capped at 2.0. High depth signals that management devoted significant discussion to the topic, consistent with genuinely new information rather than a brief mention of previously disclosed news.

Pre-Event Drift. Cumulative return over the pre-event window: `pre_drift = sum(ret_t-15, …, ret_t-1)` . Captures information already priced before the call via news or press releases. Used both as a control variable and as the basis for an interaction test (Model 4).

8-K Pre-Disclosure Indicator *(pending, future extension)*. A sharper approach to the pre-pricing problem would flag topic clusters as potentially pre-priced if the firm filed an 8-K discussing the same salience cluster in the 30 days prior to the earnings call. Using EDGAR 8-K filing dates and applying the same salience dictionary to 8-K text would allow a direct topic-level match. This extension is recognized as materially improving identification but is deferred due to the significant additional data generation required as 8-K text retrieval and NLP processing for 200k+ firm-events falls outside the current project timeline.

Adjusted Novelty Score (per turn, aggregated to firm-event). The signal construction follows a two-stage design: a precision gate followed by an additive strength score.

Stage 1: Precision gate (all three must pass):

- Novelty score ≥ 0.60, meaning the turn is semantically distant from the firm's historical baseline
- Salience score ≥ 0.30, meaning the turn contains financially material content
- LLM intent filter = YES, meaning Ollama classifies the turn as a genuine business catalyst

Turns failing any gate are excluded from the adjusted novelty score entirely. This avoids the problem of a multiplicative structure collapsing the score when any single component is near zero for reasons unrelated to signal quality (e.g., a genuinely novel, financially salient turn that no analyst happened to ask about).

Stage 2: Additive strength score (for turns passing the gate):

$$\mathrm{AdjNov} = \mathrm{novelty}\left(\frac{1}{3}, \mathrm{section_weight} + \frac{1}{3}, \mathrm{attention} + \frac{1}{3}, \mathrm{topic_depth_weight}\right)$$

The novelty score provides the base magnitude. The bracket is a weighted average of the three strength components, section weight (1.0 for prepared remarks, 1.5 for Q&A), analyst attention (fraction of unique analysts probing the topic), and topic depth (normalised chunk fraction, capped at 2.0), so the combined modifier ranges between the minimum and maximum component rather than collapsing toward zero when any one is small. Equal weights of $\frac{1}{3}$ are used as the default prior; sensitivity to alternative weightings is tested in robustness checks.

The cross-market frequency penalty is applied multiplicatively at the topic-cluster level before aggregation, as it reflects an external market-wide discount rather than a property of the individual turn:

$$\mathrm{AdjNov_{final}} = \mathrm{AdjNov} \times \mathrm{penalty}(f)$$

Aggregated as the mean across all gate-passing turns per firm-event. Firms are assigned to within-quarter quintiles (Q1 = lowest novelty, Q5 = highest).

BM25 Lexical Baseline. A rolling per-firm-quarter BM25 score (k_1 = 1.5, b = 0.75) provides a lexical benchmark for the semantic novelty measure, with built-in document-length normalization and TF saturation. In robustness tests, BM25-based novelty may be substituted for the rolling TF-IDF measure; the choice between them will be guided by which produces a more stable within-quarter novelty distribution.

4. Data Policy & Governance

4.1 Sample Selection

The sample spans April 14, 2005 through December 23, 2025. Inclusion criteria:

1. Event type must be "Earnings Calls" (homogeneous event type throughout the dataset).
2. CRSP `permno` must be non-missing (drops approximately 5,093 observations lacking CRSP linkage).
3. The firm must have at least two prior transcripts to establish a historical linguistic baseline.
4. Transcript `word_count` must exceed 100 to exclude stubs or failed transcripts.
5. All returns in the event window [−1, +5] must be non-missing.

The dataset includes firms that may have delisted during the sample period; no survivorship filtering is applied.

4.2 Missing Data Treatment

- **Missing permno (5,093 observations):** Excluded, as reliable abnormal return computation requires CRSP linkage.
- **Missing returns in the event window:** Firm-events with any missing return in [−1, +5] are dropped.
- **Missing or stub transcripts:** Observations with `word_count` < 100 are excluded as likely incomplete.

4.3 Outlier Handling

- **Returns:** Winsorized at the 1st and 99th percentiles to mitigate microstructure noise and data errors.
- **Novelty scores:** No winsorization, as scores are bounded in [0, 1] by construction.
- **Transcript length:** The distribution is documented in the EDA (Section 5); no trimming is applied unless extreme outliers distort NLP measures.

4.4 Look-Ahead Bias Prevention

- **Rolling baseline:** For each firm-event, the embedding centroid, TF-IDF vocabulary, and BM25 baseline use only transcripts from strictly prior quarters within the rolling window. No future information enters the novelty computation. The rolling window specification (trailing four quarters or exponential decay) is selected before empirical analysis and held fixed across all models.
- **Cross-market frequency penalty:** Computed with the trailing three month window to avoid lookahead bias.

- **Return alignment:** Pre-computed returns are aligned such that `ret_t0` corresponds to the event date. The close-to-open return uses the call-day close and next-day open which both are post-event observations, consistent with how the market processes the information.

5. Exploratory Data Analysis

5.1 Summary Statistics

Table 1 - Panel A: Financial Variables - Summary Statistics

Variable	N	Mean	Std	p1	p25	p50	p75	p99
ret_t=0	185,622	0.0004	0.0699	−0.200	−0.026	0.000	0.027	0.196
ret_t=+1	185,628	−0.0010	0.0741	−0.224	−0.025	0.000	0.023	0.217
Close-to-Open	185,588	0.0002	0.0518	−0.176	−0.009	0.000	0.011	0.159

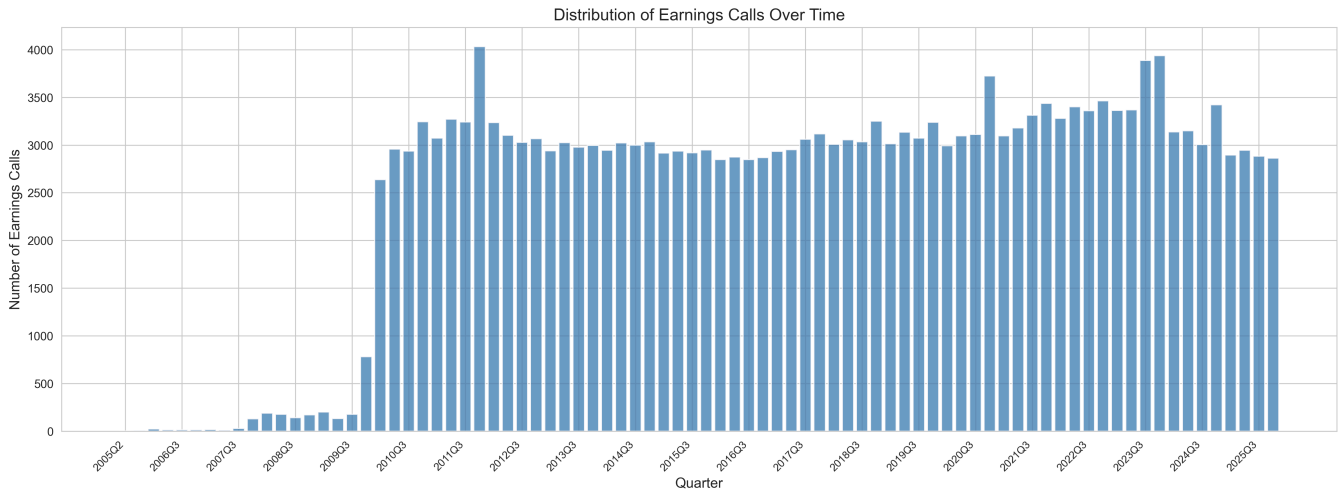
Returns are winsorized at the 1st and 99th percentiles. The event-day return (`ret_t0`) exhibits roughly double the standard deviation of non-event days (σ = 0.070 vs. ≈ 0.033), confirming that earnings calls are high-information events. The mean event-day return is near zero (0.04 bps), while the next-day return is slightly negative (−0.10 bps), consistent with a modest mean reversion pattern.

Table 2 - Panel B: Text Variables - Summary Statistics

Variable	N	Mean	Std	p1	p25	p50	p75	p99
Word Count	202,357	9,624	6,457	2,040	5,760	8,089	11,127	33,380
Transcript Length (chars)	202,357	56,275	37,491	12,349	33,893	47,229	64,975	194,245

The median transcript contains 8,089 words (~47,000 characters). The distribution is right-skewed (mean > median), with the 99th percentile reaching 33,380 words, reflecting occasional extended calls with lengthy Q&A sessions.

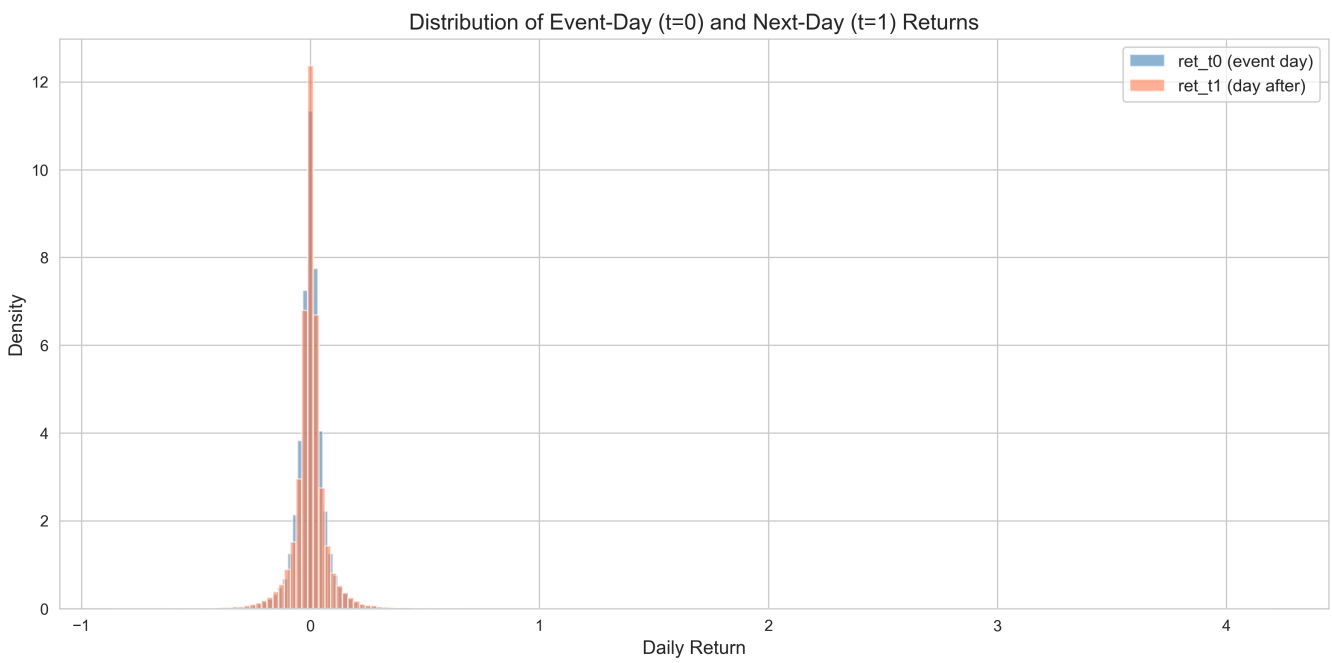
Figure 1: Earnings Calls per Calendar Quarter, 2005–2025



Number of earnings calls per calendar quarter, 2005–2025. Coverage begins with fewer than 200 calls per quarter in 2005–2008, then ramps sharply to approximately 2,700–3,000 by 2010. From 2010 onward, quarterly volume stabilizes at roughly 2,900–3,300 calls, with periodic peaks near 4,000 (2011Q2, 2020Q3, 2023Q3–Q4). A notable spike in 2020Q3 reflects the wave of COVID-related earnings updates. The 2025 quarters show a slight decline, likely due to incomplete data coverage at the end of the sample.

5.2 Return Distributions

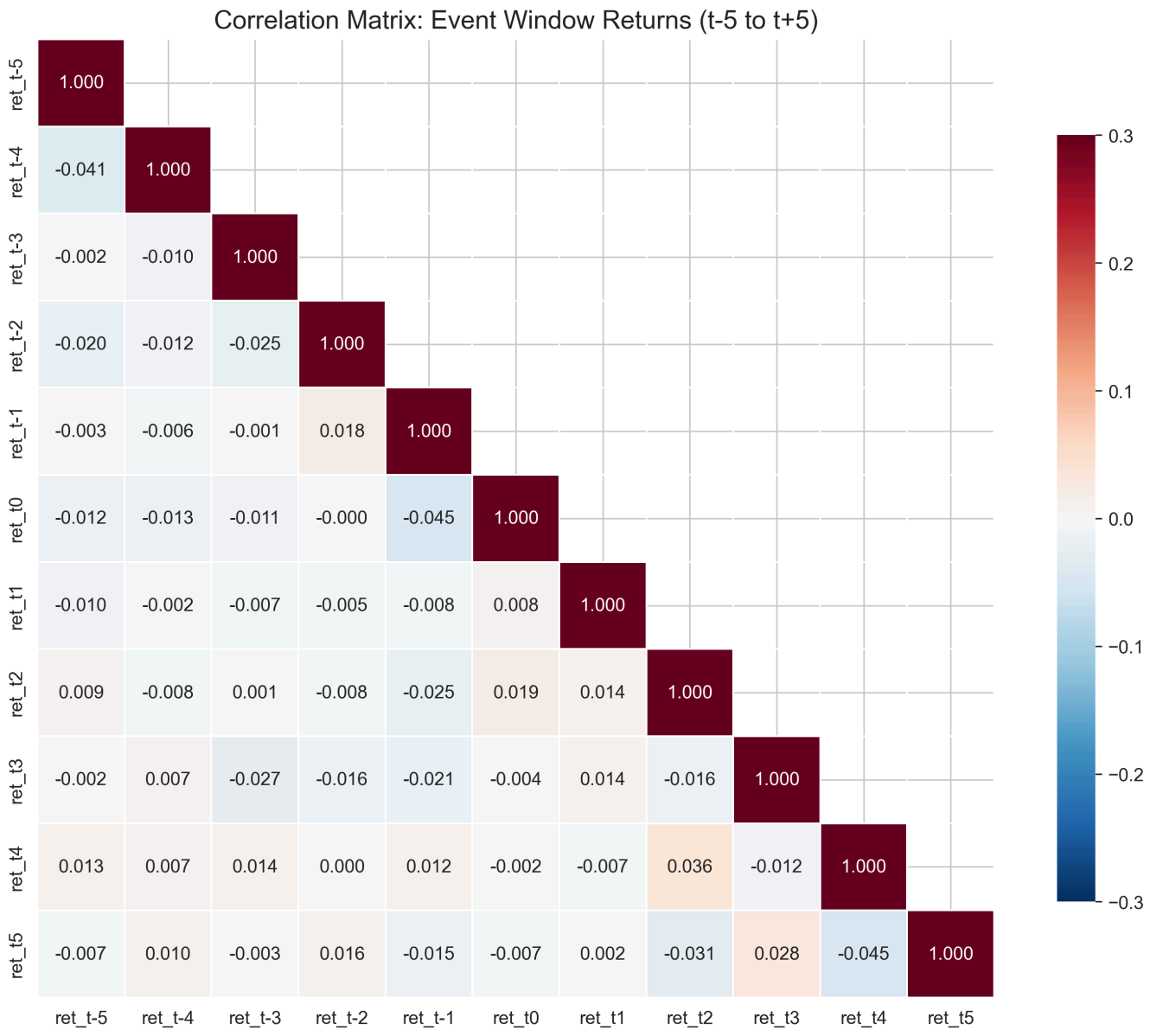
Figure 2: Event-Day and Next-Day Return Distributions



Overlaid density histograms of event-day return (`ret_t0` , blue) and next-day return (`ret_t1` , orange). Both distributions are sharply peaked near zero with heavy tails, but `ret_t0` exhibits a wider spread (σ = 0.070) than typical non-event days (σ ≈ 0.033), confirming the information content of earnings calls. Extreme positive outliers extend

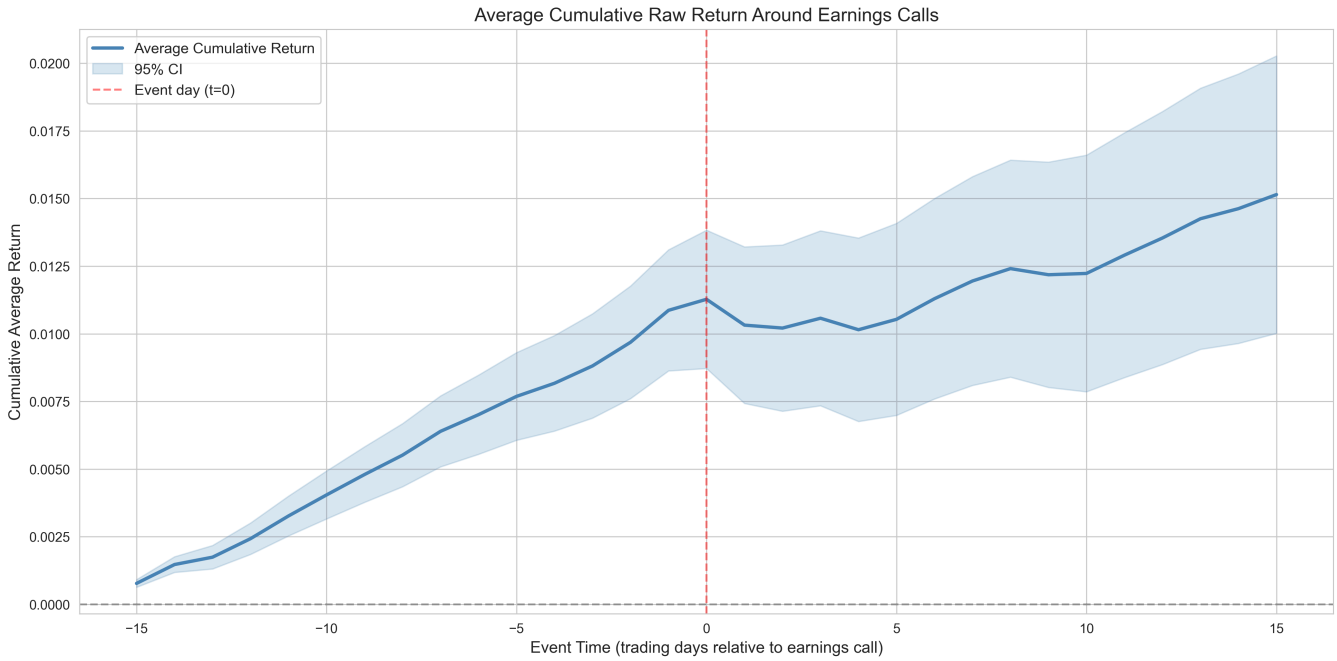
beyond +300%, reflecting occasional transformative announcements. The next-day distribution (`ret_t1`) shows a slightly higher peak density near zero and comparable tail behavior, consistent with continued price adjustment following the event.

Figure 3: Return Correlation Heatmap



Pairwise correlation matrix of daily returns from t−5 through t+5. Off-diagonal correlations are uniformly small in magnitude (|r| < 0.05), confirming low serial dependence in daily returns. The most notable off-diagonal entry is the negative correlation between ret_t−1 and ret_t0 (r = −0.045) and between ret_t4 and ret_t5 (r = −0.045), suggesting mild mean reversion at adjacent horizons. The event-day return (ret_t0) shows near-zero correlation with all pre-event returns, consistent with the semi-strong form of market efficiency. Cross-correlations between post-event days (t+1 through t+5) are also small, indicating that post-earnings adjustment occurs primarily on the event day itself.

Figure 4: Average Cumulative Return Around Event Date



Average cumulative raw return from t−15 through t+15, computed across all 185,622 firm-events with 95% confidence interval. The trajectory shows a steady upward pre-event drift from approximately 0 to 100 bps over the 15 days before the call, consistent with positive information leakage or selection. A small discrete uptick is visible at t=0 (the event day). Post-event, cumulative returns plateau briefly around 100 bps at t+1 through t+5 before resuming an upward drift to approximately 150 bps by t+15. The absence of a sharp drop after t=0 suggests that, on average across all firms, there is no aggregate mean reversion. The reversion effect, if present, is specific to the high-novelty quintile and is masked in the unconditional average.

6. Portfolio Construction

The primary empirical test follows the calendar-time portfolio methodology of Cohen, Malloy & Nguyen (2020), enabling direct comparison of our results to their 10-K language change strategy. At the end of each calendar quarter, all firms with valid adjusted novelty scores are ranked within-quarter into quintiles (Q1 = lowest novelty, Q5 = highest). A long-short portfolio is formed by buying Q1 (familiar language, expected underreaction) and shorting Q5 (novel language, expected overreaction). Positions are held for one quarter and rebalanced at each subsequent ranking date.

6.1 Portfolio Formation

- **Ranking period:** End of quarter t, using `AdjNov` computed from the current quarter's earnings call transcripts.
- **Holding period:** Quarter t+1 (one quarter, consistent with Cohen et al.'s one-year holding period scaled to quarterly rebalancing given the earnings call frequency).
- **Weighting:** Equal-weighted within each quintile. Value-weighted versions are run as robustness checks.

- **Short-selling:** The strategy assumes frictionless shorting. Shorting costs are acknowledged as a limitation; in practice the short leg would be implemented via put options or index hedging for high-cost-to-borrow names.

6.2 Return Computation

Monthly portfolio returns are computed for each quintile:

$$R_{p,\tau}^{(q)} = \frac{1}{N_\tau^{(q)}} \sum_{i \in q} R_{i,\tau}$$

The long-short spread is:

$$R_{LS,\tau} = R_{p,\tau}^{(Q1)} - R_{p,\tau}^{(Q5)}$$

6.3 Risk Adjustment

Alpha is estimated by regressing monthly long-short returns on the Fama-French five-factor model:

$$R_{LS,\tau} - R_{f,\tau} = \alpha + \beta_1 MKT_\tau + \beta_2 SMB_\tau + \beta_3 HML_\tau + \beta_4 RMW_\tau + \beta_5 CMA_\tau + \varepsilon_\tau$$

The intercept α is the novelty-strategy alpha net of standard risk exposures. Results are benchmarked against Cohen et al.'s reported 188 bps/month alpha; any positive and significant α in our specification is interpreted as consistent with the overreaction mechanism.

6.4 Event-Window Supplement

Alongside the portfolio sort, cumulative abnormal returns (CARs) are computed around each earnings call date to characterize the short-window return pattern by quintile. This supplements but does not replace the portfolio analysis. The event date (τ = 0) is `mostimportantdateutc`. Windows examined:

- **Estimation window:** [t-15, t-2], a 14-trading-day window for expected return estimation (constant mean model).
- **Event window:** [t-1, t+1], capturing the immediate market reaction.
- **Post-event window:** [t+2, t+5], used for the mean reversion test.
- **Extended window:** [t+2, t+15], covering the full drift analysis.

Expected return (constant mean model):

$$\bar{R}_i = \frac{1}{14} \sum_{\tau=-15}^{-2} R_{i,\tau}$$

Abnormal return:

$$AR_{i,\tau} = R_{i,\tau} - \bar{R}_i$$

Cumulative abnormal return:

$$CAR_i(\tau_1, \tau_2) = \sum_{\tau=\tau_1}^{\tau_2} AR_{i,\tau}$$

Average abnormal return across firms in quintile q:

$$AAR_\tau^{(q)} = \frac{1}{N_\tau^{(q)}} \sum_{i \in q} AR_{i,\tau}$$

$$CAAR(\tau_1, \tau_2) = \sum_{\tau=\tau_1}^{\tau_2} AAR_\tau$$

7. Model Specification

7.1 Dependent and Independent Variables

Dependent variables:

- $CAR_i[-1, +1]$: Cumulative abnormal return over the 3-day event window (primary). Captures the full immediate market reaction including any pre-event leakage.
- $CAR_i[+2, +5]$: Cumulative abnormal return over the post-event window (for mean reversion tests).

Key independent variables:

- $NovScore_i$: Adjusted novelty score (continuous), aggregated to the firm-event level.
- $Q2_i, Q3_i, Q4_i, Q5_i$: Within-quarter novelty quintile dummies (Q1 = lowest novelty, omitted reference group).
- $close\ to\ open_i$: Overnight return proxy for earnings surprise direction and magnitude.
- SUE_i: Standardized unexpected earnings from I/B/E/S, defined as (actual EPS − consensus EPS) / std(analyst forecasts). The critical control for separating the language signal from the fundamental earnings signal.
- $pre\text{-}drift_i$: Cumulative pre-event return, capturing information already priced before the call.

7.2 Section Weighting & Analyst Attention

The novelty score incorporates two layers of weighting that reflect the economics of information revelation during earnings calls.

Section weighting assigns 1.0 to prepared remarks and 1.5 to Q&A management responses. The economic rationale is that prepared remarks are pre-scripted and heavily vetted by legal and investor relations teams, whereas Q&A responses are elicited by analyst questions and are less controllable by management. Novelty in the Q&A section, particularly on topics that management did not voluntarily address is a stronger signal of material, unplanned disclosure.

Analyst attention scoring measures the fraction of unique analysts who ask about each topic cluster in the Q&A section. Topics that attract questions from multiple analysts represent consensus surprise among sophisticated market participants. The attention score serves as a revealed-preference filter: topics that only one analyst finds interesting are downweighted relative to topics that multiple analysts probe independently.

7.3 Pre-Event Drift & Pre-Priced Novelty

A key identification challenge is that content flagged as "novel" by the NLP pipeline may already be known to the market. Between earnings calls, firms issue press releases, file 8-Ks, participate in investor conferences, and receive analyst coverage. A new product launch announced via press release three weeks before the earnings call will appear as "novel" in the transcript (never discussed in prior calls) even though the market has already priced the information.

We address this through multiple mechanisms:

1. **Pre-event drift control:** `pre_drift = sum(ret_t-15, …, ret_t-1)` captures the cumulative price adjustment before the call. A large absolute pre-drift suggests that significant information has already been incorporated.
2. **Model 4 interaction term:** `NovScore` × | `pre_drift` | tests whether the novelty effect is attenuated when pre-drift is large. A negative interaction coefficient supports the pre-pricing hypothesis.
3. **Topic depth filter:** Topics mentioned briefly (low depth) are more likely to be "old news" being acknowledged in the transcript; topics discussed at length (high depth) are more likely to contain genuinely new details.
4. **Analyst attention filter:** If a topic was already disclosed and priced, analysts are less likely to ask about it, reducing its attention weight.

7.4 Control Variables

Table 3: Control Variables

Variable	Construction	Rationale
`SUE`	(Actual EPS − Consensus EPS) / σ(forecasts), from I/B/E/S	Separates language signal from fundamental earnings signal; most critical control
`close_to_open`	Overnight return (call-day close → next open)	Proxy for immediate earnings surprise magnitude and direction
`log_mktcap`	log(market cap) from CRSP, prior quarter-end	Size effect; larger firms have more analyst coverage and less mispricing
`book_to_market`	Book equity / market equity from Compustat/CRSP	Value premium control; novelty signal could proxy for growth vs. value
`analyst_coverage`	Number of analysts with forecasts on I/B/E/S	Attention control; low-coverage firms mechanically score as novel due to thinner language standardization
`log_word_count`	log(word_count)	Controls for transcript length
`pre_drift`	Σ `ret_t-15` through `ret_t-1`	Info already priced before call
`qa_ratio`	Q&A words / total words	Controls for call structure
Quarter FE	Calendar quarter indicator	Market-wide trends, macro conditions
Year FE	Calendar year indicator	Secular trends in efficiency

Standard errors are clustered by calendar date throughout all regressions, following the recommendations for event studies with cross-sectionally correlated events.

7.5 Regression Equations (Models 1–4)

Model 1 (Quintile-based, main specification):

$$CAR_i[-1,+1] = \alpha + \beta_1 Q2_i + \beta_2 Q3_i + \beta_3 Q4_i + \beta_4 Q5_i + \gamma_1 \,\mathrm{SUE}_i + \gamma_2 \,\mathrm{close\ to\ open}_i + \gamma_3 \log(\mathrm{mktcap}_i$$

The quintile dummies allow a non-parametric relationship between novelty and returns. Under H1, we expect β_4 (Q5) to be significantly different from zero and larger in magnitude than β_1 through β_3.

Model 2 (Continuous novelty with quadratic term):

$$CAR_i[-1,+1] = \alpha + \beta_1 \,\mathrm{NovScore}_i + \beta_2 \,\mathrm{NovScore}_i^2 + \gamma_1 \,\mathrm{SUE}_i + \gamma_2 \,\mathrm{close\ to\ open}_i + \gamma_3 \log(\mathrm{mktcap}_i) +$$

The quadratic term tests for a non-linear (potentially U-shaped or inverted-U) relationship between novelty and abnormal returns.

Model 3 (Mean reversion test):

$$CAR_i[+2,+5] = \alpha + \beta_1 Q5_i + \beta_2 \, CAR_i[-1,+1] + \beta_3 \big(Q5_i \times CAR_i[-1,+1]\big) + \gamma_1 \,\mathrm{SUE}_i + \gamma_2 \log(\mathrm{mktca}$$

The interaction Q5 × CAR[−1,+1] is the key test of H2. A negative β_3 indicates that high-novelty firms with large initial reactions experience stronger reversals, consistent with overreaction followed by correction.

Model 4 (Pre-priced novelty test):

$$CAR_i[-1,+1] = \alpha + \beta_1 \,\mathrm{NovScore}_i + \beta_2 \,|\mathrm{pre\text{-}drift}_i| + \beta_3 \big(\mathrm{NovScore}_i \times |\mathrm{pre\text{-}drift}_i|\big) + \gamma_1 \,\mathrm{SUE}_i + \gamma_2 \,\mathrm{clos}$$

A negative β_3 supports the pre-pricing hypothesis: when the market has already moved substantially before the call (large |pre_drift|), the novelty in the transcript has less incremental price impact.

7.6 Identification Strategy

Several identification challenges must be acknowledged:

Omitted variable bias. Novelty may correlate with firm characteristics such as size, growth, and industry that independently predict returns. Quarter and year fixed effects partially absorb time-varying confounders. Firm-level controls for market capitalization, book-to-market ratio, and analyst coverage are included in all specifications to address this concern directly.

Simultaneity. Novelty is measured from the same event that generates the returns. The rolling historical baseline mitigates this concern: novelty is measured as deviation from the firm's own prior language, not from contemporaneous market conditions. The text is produced before the market reacts, and the embedding pipeline uses only pre-event information for the baseline.

Selection. Firms that discuss novel topics may differ systematically from those that do not. Importantly, our pipeline scores *all* firms in the sample (not just those exceeding some novelty threshold), avoiding selection on the treatment

variable. The quintile assignment is cross-sectional within each quarter, so even low-novelty firms contribute to the analysis.

Pre-priced novelty. Addressed through the multi-layered approach described in Section 7.3, pre-drift controls, interaction tests, topic depth, and analyst attention.

Cross-sectional dependence. Standard errors clustered by calendar date account for the correlation among firms reporting on the same day, following standard practice in the event study literature.

Appendix A. Salience Dictionary Clusters

The salience scoring module (`salience_dictionary.py`) defines 26 weighted clusters. Each cluster has a `cluster_weight` (1.5–3.0) reflecting its typical valuation impact, and each term has an individual weight (0.4–1.0) reflecting specificity. Multi-word terms are matched longest-first so that "net interest margin" consumes the span before "margin" can fire separately. The `mgmt_signal` cluster is unique in containing *negative* weights for boilerplate phrases that dilute signal quality.

Table A.1: Salience Dictionary Cluster Summary and Sample Terms

Cluster	Weight	Sample High-Signal Terms (weight)	Sample Generic Terms (weight)
`guidance`	3.0	"raising guidance" (1.0), "withdrawing guidance" (1.0), "guidance range" (1.0)	"outlook" (0.9), "expect" (0.5), "going forward" (0.5)
`earnings_performance`	3.0	"beat expectations" (1.0), "missed expectations" (1.0), "record quarter" (0.9)	"eps" (0.7), "net income" (0.7), "exceeded" (0.7)
`revenue_growth`	2.5	"organic growth" (1.0), "same-store sales" (1.0), "remaining performance obligation" (1.0)	"market share" (1.0), "bookings" (0.9), "pipeline" (0.6)
`margin`	2.2	"gross margin expansion" (1.0), "operating leverage" (1.0), "margin compression" (1.0)	"gross margin" (0.9), "cost of goods" (0.8), "pricing power" (0.8)
`capital_alloc`	2.5	"share repurchase program" (1.0), "special dividend" (1.0), "buyback authorization" (1.0)	"dividends" (0.8), "capex" (0.7), "return on capital" (0.8)
`balance_sheet`	2.5	"debt covenant" (1.0), "leverage ratio" (1.0), "credit facility" (0.9)	"long-term debt" (0.8), "liquidity" (0.7), "cash position" (0.7)
`product_innov`	2.2	"product launch" (1.0), "new platform" (0.9), "patent" (0.9)	"roadmap" (0.7), "r&d" (0.6), "pipeline" (0.6)

Cluster	Weight	Sample High-Signal Terms (weight)	Sample Generic Terms (weight)
`partnerships`	2.0	"strategic partnership" (1.0), "joint venture" (1.0), "licensing agreement" (0.9)	"collaboration" (0.7), "agreement" (0.6), "partnership" (0.7)
`competition`	2.2	"market share loss" (1.0), "competitive displacement" (1.0), "pricing pressure" (0.9)	"competitors" (0.7), "competitive" (0.6), "differentiation" (0.6)
`macro_env`	2.5	"recession risk" (1.0), "rate sensitivity" (0.9), "fx headwind" (0.9)	"inflation" (0.8), "interest rates" (0.8), "uncertainty" (0.7)
`supply_chain`	2.2	"supply disruption" (1.0), "inventory build" (0.9), "lead times" (0.9)	"supply chain" (0.8), "logistics" (0.6), "inventory" (0.6)
`regulatory`	1.8	"regulatory approval" (1.0), "compliance penalty" (1.0), "sec investigation" (1.0)	"regulatory" (0.7), "compliance" (0.6), "litigation" (0.7)
`technology`	2.2	"ai integration" (1.0), "cloud migration" (1.0), "digital transformation" (0.9)	"technology" (0.6), "platform" (0.6), "software" (0.6)
`geo_expansion`	2.0	"market entry" (1.0), "international expansion" (0.9), "emerging markets" (0.9)	"international" (0.6), "global" (0.5), "geographic" (0.6)
`workforce`	1.8	"restructuring charge" (1.0), "headcount reduction" (1.0), "talent acquisition" (0.8)	"employees" (0.5), "hiring" (0.5), "workforce" (0.6)
`banking_credit`	1.5	"net interest margin" (1.0), "loan loss provision" (1.0), "nonperforming loans" (1.0)	"deposits" (0.7), "credit quality" (0.8), "lending" (0.6)
`healthcare`	1.5	"fda approval" (1.0), "clinical trial" (1.0), "drug launch" (1.0)	"patients" (0.6), "treatment" (0.6), "pipeline" (0.6)
`energy`	1.5	"proved reserves" (1.0), "refining margin" (1.0), "lng capacity" (0.9)	"oil price" (0.7), "production" (0.6), "upstream" (0.7)
`customer_metrics`	2.0	"net revenue retention" (1.0), "customer churn" (1.0), "annual recurring revenue" (1.0)	"customers" (0.5), "retention" (0.7), "attach rate" (0.8)
`pricing_strategy`	2.0	"price increase" (0.9), "yield management" (0.9), "pricing realization" (0.9)	"pricing" (0.6), "price" (0.4), "volume" (0.5)

Cluster	Weight	Sample High-Signal Terms (weight)	Sample Generic Terms (weight)
`esg`	1.8	"carbon neutral commitment" (1.0), "sustainability target" (0.9), "scope 3 emissions" (0.9)	"sustainability" (0.7), "esg" (0.7), "environmental" (0.6)
`consumer_demand`	2.0	"consumer confidence" (0.9), "discretionary spending" (0.9), "wallet share" (0.9)	"demand" (0.5), "consumer" (0.5), "spending" (0.5)
`real_estate`	1.5	"cap rate compression" (1.0), "occupancy rate" (1.0), "same-store noi" (1.0)	"occupancy" (0.7), "leasing" (0.7), "rental" (0.6)
`retail`	1.5	"comparable store sales" (1.0), "store traffic" (0.9), "inventory turnover" (0.9)	"same-store" (0.9), "foot traffic" (0.8), "shrink" (0.7)
`mgmt_signal`	1.8	*(negative weights)* "as I mentioned" (−0.3), "going forward" used generically (−0.2)	"pleased to report" (−0.4), "thank you operator" (−0.5)
`segment_reporting`	1.8	"segment margin" (0.9), "business unit performance" (0.9), "divisional revenue" (0.8)	"segment" (0.5), "division" (0.5), "unit" (0.4)

Cluster weights reflect typical valuation materiality (3.0 = highest). Term weights reflect specificity within a cluster (1.0 = highly specific signal, 0.4 = generic, only fires when no better term matches). The `mgmt_signal` cluster uses negative weights to penalise boilerplate phrases that dilute financial signal across other clusters.

Appendix B. NLP Pipeline Parameters

Table B.1: NLP Pipeline Parameter Settings

Parameter	Value	Rationale
Semantic encoder	`nomic-embed-text-v1.5` (768-dim)	Purpose-built for semantic similarity and retrieval; embedding space preserves meaning, not sentiment polarity
LLM intent filter	Ollama (local, e.g. `llama3.1:8b`)	Binary YES/NO business catalyst classification; locally hosted, no API cost, model-swappable
Lexical baseline A	Firm-level rolling TF-IDF	Per-firm vocabulary profile; closest analog to Cohen et al. (2020)

Parameter	Value	Rationale
Lexical baseline B	3-month universe TF-IDF	Cross-sectional novelty relative to contemporaneous market language
Lexical baseline C	BM25 (k_1 = 1.5, b = 0.75)	Lexical benchmark with TF saturation and doc-length normalization
Min. prior transcripts	2	Sufficient history to define "normal" within rolling window
Rolling window	Trailing 4 quarters (primary); exponential decay (robustness)	Prevents dilution of baseline by structurally distant historical periods
Novelty threshold	0.60	turns above threshold are candidates for LLM intent filter
Salience threshold	0.30	Minimum financial relevance for inclusion in adjusted score
LLM pre-filter threshold	novelty ≥ 0.65, salience ≥ 0.50	Only highest-confidence turns sent to Ollama to minimise latency
Macro penalty	sigmoid, f_0 = 0.40, k=15	Smooth decay around 40% cross-firm frequency; eliminates hard cliff
Section weight (prepared)	1.0	Baseline
Section weight (Q&A)	1.5	Analyst-elicited, less scripted
Analyst attention similarity	0.75 cosine	Topic cluster formation threshold
Topic depth cap	2.0	Prevents single-topic dominance

Appendix C. Variable Definitions

Table C.1: Variable Definitions

Variable	Definition
`CAR[-1,+1]`	Cumulative abnormal return, event window
`CAR[+2,+5]`	Cumulative abnormal return, post-event window
`R_LS`	Monthly long-short portfolio return: Q1 minus Q5

Variable	Definition
`alpha`	Intercept from FF5 regression of R_LS on market factors
`NovScore`	Adjusted novelty score (firm-event level), nomic-embed-text-v1.5
`Q1 – Q5`	Within-quarter novelty quintile dummies
`SUE`	Standardized unexpected earnings: (actual − consensus EPS) / σ(forecasts), I/B/E/S
`close_to_open`	Overnight return (earnings surprise proxy)
`log_mktcap`	Log of market capitalisation from CRSP, prior quarter-end
`book_to_market`	Book equity / market equity from Compustat/CRSP, prior quarter-end
`analyst_coverage`	Number of analysts with active forecasts on I/B/E/S at event date
`log_word_count`	Log transcript word count
`pre_drift`	Cumulative return over [t−15, t−1]
`qa_ratio`	Q&A section word count / total word count
`section_weight`	1.0 (prepared remarks) or 1.5 (Q&A management turns)
`attention_score`	Fraction of unique analysts asking about topic cluster
`topic_depth_weight`	Turn fraction per topic, normalized by median, capped at 2.0
`llm_intent_pass`	Binary: 1 if Ollama classifies turn as genuine business catalyst
`bm25_novelty`	BM25-based lexical novelty score (robustness baseline)
`tfidf_firm_novelty`	Firm-level rolling TF-IDF novelty score (lexical baseline A)
`tfidf_univ_novelty`	3-month universe TF-IDF novelty score (lexical baseline B)

Appendix D. Sample Construction Flow

Table D.1: Sample Construction Flow

Step	Observations
Starting sample	202,357
Less: missing permno	(5,093)
Less: < 2 prior transcripts	(TBD)
Less: word_count < 100	(TBD)
Less: missing event returns	(TBD)

Step	Observations
Final analysis sample	**~TBD**